Jerry (Yuze) Li

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Education

University of Waterloo Expected 2029
Bachelor of Computer Science | Schulich Leader Scholarship ($100,000+) *Waterloo, Canada*

Experience

Waterloo Defence *Software Lead* Sept. 2026 – Present
- Lead software for a student team's autonomous counter-drone turret, targeting the 2027 DND CUAS Urban Sandbox.
- Build camera detection and tracking of small drones, a ballistic solver for lead, drag, and drop, and a MuJoCo sim.

Mathos AI (YC W24) *Software Engineer Intern* June 2026 – Present
- Shipped MP4 export for AI math videos (Remotion on AWS Fargate via Celery) and SES/EventBridge lifecycle emails.
- Built an OAuth 2.0 and MCP server so ChatGPT and Claude users can generate and download Mathos videos in chat.
- Rebuilt the AI tutor's step grading on a fast judge model plus SymPy checks, sharply improving accuracy and speed.

TopsOJ *Founder & Lead Developer* July 2023 – Present
- Built a competitive-math judge with 100,000+ users in 180+ countries (Google Analytics); led a 30-person team.

Quoted Tech Computers *Software Engineer Intern* July 2025 – Aug. 2025
- Automated barcode-label printing (Node.js), cutting manual processing time by 70%; built Playwright catalog scrapers.

Research

AI Agents 2026
Sole author; accepted at two NeurIPS 2026 workshops *Sydney*
- "Verifying the Second Agent: Controlled Experiments on Recursive Delegation in Coding Agents." In two within-task crossover campaigns on verifier-scored repository repairs, recursive agent trees spent 6× to 12× the tokens of a single agent and no contrast resolved a gain in verified fixes; 90% of refused sibling patches collided at the line level.

Computational Astrophysics Nov. 2024 – Mar. 2026
Sole author of three papers, supervised by Dr. Alexander Mushtukov (University of Oxford) *Remote*
- "Ballistic Mass Transfer Necessarily Contracts Orbits in Binary Star Systems," *Research Notes of the AAS* (2026).
- "On the Conditions Under Which Instantaneous Mass-Transfer Mappings Reproduce Conservative Limits," *Journal of High School Science* (2026), **Best Manuscript**. A velocity-rescaling closure cut orbit error from 22% to under 10^{-4}%.
- "The Effective Velocity of Transferred Mass," arXiv:2608.19107, under review at *Astronomische Nachrichten*.

Projects

Cardboard Cockpit | *Godot, Python/OpenCV, ArUco, WebSockets, BLE* Hack the North 2026
- Built most of the Godot flight sim (21 of 22 PRs): map, weapons, HUD, BLE badge cockpit, landing, and tutorial.

Daedalus | *TypeScript, Three.js/WebGL, MediaPipe* JAMHacks 10
- Co-built a webcam-gesture 3D modeling app with MediaPipe hand tracking and Three.js; won Best Non-GenAI Hack.

Drift | *JavaScript, Chrome Extensions API* 2026
- Built a Chrome extension that opens a focused feed while ChatGPT, Claude, or Gemini think; Web Store Featured.

Worldline Engine | *C++, raylib, CMake* 2026
- Wrote a deterministic C++ universe simulator with seeded generation, live physics, save/replay, and automated tests.

Awards

USACO Platinum (1,000/1,000 on Gold) | **AIME** qualifier (3×)
AAPT PhysicsBowl Div. II: 1st in Canada, 7th globally | **Sir Isaac Newton Exam**: 4th in Canada, 9th globally

Technical Skills

Languages: C/C++, Python, TypeScript, JavaScript, SQL | **Tools:** AWS (Fargate, SES, S3), Git, Linux, SQLite
Frameworks: Node.js, Express, React, Next.js, Celery, NumPy, SciPy, Godot, raylib, Three.js, OpenCV, MediaPipe